

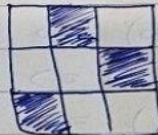
Image

- Image data quyidagi ko'rinishlarida bo'ladi:
 - Photo, Medical scan, Satellite image, CCTV frame, selfie va hokazolar
- Computerlar image larini infonlarda u fargli ko'radi. Oddiy image belatida rucos ulorui sahar jadvali shaklida ko'radi. (Qanday sonlar?)

Pixel

- Pasmning eng kichik b.r qismi.
- Har bir rasu juda ko'plab (pixel) lardan tashkil topadi.

Matblau:



Har bir kvadrat

(1 pixel)

- => Har bir pixel o'z rangi (soniga) ga bo'ladi.
uning cheporati (0, 255).
kichik son => qora
katta son => oq.

Grayscale images

- o dan 255 gacha jami 256 ta qiymatlar.
- ko'rinishi - qoradan oqgacha barcha kulrang tushari.

Binary images

0 va 1 yoki oq va 255.
faqat oq va qora rangdan iborat

Resolution

- Rasmdagi umumiy (pixellar son)
- Katta resolution $\Rightarrow$ ko'proq ma'lumot
- Kichik resolution $\Rightarrow$ kam ma'lumot

$28 \times 28 \Rightarrow 28 \text{ pixelga } 28 \text{ pixel imajimizning}$
umumiy pixellar soni $\Rightarrow$ (784 pixel)

RGB $\Rightarrow$ Red Green Blue

Har yuzidagi barcha ranglar shu
3 ta ranglar aralashmasidan hosil
bo'ladi.

RGB (255, 0, 0) $\Rightarrow$ Red

RGB (0, 255, 0) $\Rightarrow$ Green

RGB (0, 0, 255) $\Rightarrow$ Blue

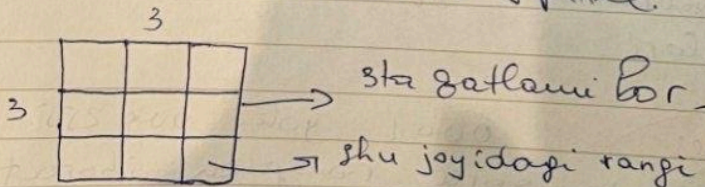
RGB (0, 0, 0) $\Rightarrow$ Black

RGB (255, 255, 255) $\Rightarrow$ White

RGB channels

Har bir rangning ustma ust qo'yilishi. Qattiq
RGB imajlarda 3 ta qatlami bo'ladi.
shu 3 ta rangdan iborat qatlami.

Masalan: 3×3 imaj bor. jami 9 pixel
resolution $\Rightarrow$ 9 pixel.



RGB imaj.

Har bir rangdan
olinilishi.

R G B
(255, 50, 200)

birlik

birlik

birlik

gacha ulab olinadi

GrayScale $\Rightarrow$ ~~Height~~ Height $\times$ Width $\times 1$
channels $\Rightarrow$ 1 so temp
 $28 \times 28 \Rightarrow 784$ pixel $\Rightarrow 784$ son.

$H \times W$

RGB image $\Rightarrow$ Height $\times$ Width $\times 3$
 $H \times W \times 3$

$28 \times 28 \Rightarrow 784$ pixel.

$28 \times 28 \times 3 \Rightarrow 2352$ to son.

Transparency

Image da zancholik pixelor
korinikliini nazorat qiladi.

RGB $\rightarrow$ RGBA (alpha).

$A \Rightarrow 0$ butunlay shaffof (pixel korinmaydi).

$A \Rightarrow 255$ butunlay zica (fon umuman korinmaydi).

$A \Rightarrow 128$ yarim shaffof.

RGBA images channels 4 ta bo'lib qoladi.
 $H \times W \times 4$

$28 \times 28 \times 4 \Rightarrow$ pixelar soni - 784.

Hov pixelole 4 ta son (R, G, B, A)

jami sonlar 3136.